

Cross-Asset Correlation

When Everything Moves Together — Lesson Explainer

The Concept

Stocks, bonds, currencies, and commodities each have typical correlation relationships, but those relationships aren't fixed - they shift depending on the market regime (calm versus crisis).

In calm markets, stocks and government bonds often show low or negative correlation - genuine diversification. In a genuine crisis, a broad 'risk-off' panic can cause investors to sell everything simultaneously, pushing correlations across many asset classes sharply toward +1.

This makes cross-asset correlation a real risk-management concern: a portfolio that looks diversified based on calm-market history can behave like a concentrated bet exactly when a crisis hits.

Illustrative Example: Correlation Across Two Regimes

Asset Pair	Calm-Regime Corr.	Crisis-Regime Corr.
Stocks vs. Government Bonds	-0.30	+0.55
Stocks vs. Corporate Bonds	+0.40	+0.85
Stocks vs. Commodities	+0.15	+0.60
Stocks vs. Safe-Haven Currency	-0.20	-0.65

Most pairs move toward +1 in the crisis regime, except the safe-haven currency, whose negative correlation to stocks actually strengthens in a crisis - exactly what makes it valuable as a hedge when it's needed most.

5 Things to Know About Cross-Asset Correlation

1. Correlations are regime-dependent, not fixed - relationships can reverse or intensify.
2. Most assets correlate upward in a crisis - a sell-everything dynamic pushes toward +1.
3. Not every asset behaves the same way - some maintain or strengthen negative correlation.
4. Historical correlation is a guide, not a guarantee - values shift across different crisis types.
5. This is why stress-testing matters - evaluate a portfolio under both calm and crisis assumptions.

4 Things to Check When Assessing Cross-Asset Risk

- Check correlations across regimes - look at both calm-period and stress-period history.
- Identify genuine crisis diversifiers - assets that held up under stress historically.
- Stress-test the full portfolio - model it under a crisis-correlation scenario, not just calm.
- Track cross-asset signals for regime shifts - rising correlation can itself be an early sign.

Worth knowing:

this explains cross-asset correlation dynamics using illustrative numbers - it isn't personalized financial advice, and no correlation value or diversification claim here is a recommendation for your own portfolio. Correlations shift over time and no asset is guaranteed to behave the same way in every future stress event.

Quiz — Answer the questions, then check the key on the next page

1. What does this lesson mean by correlations being "regime-dependent"?

- A. Correlations are set by government regulation
- B. Correlations are the same in every market condition
- C. Correlation relationships between assets can shift depending on whether markets are calm or stressed
- D. Correlations only apply to cryptocurrency markets

2. What typically happens to most cross-asset correlations during a genuine crisis?

- A. They all drop to exactly zero
- B. They tend to rise toward +1, as a broad sell-everything dynamic takes hold
- C. They become impossible to measure
- D. They reverse to become perfectly negative for every pair

3. In the illustrative example, what was notable about the safe-haven currency's correlation to stocks?

- A. It became positive during the crisis, like most other pairs
- B. It stayed at exactly zero in both regimes
- C. It was impossible to calculate
- D. Its negative correlation to stocks actually strengthened during the crisis

4. Why is cross-asset correlation described as a real risk-management concern?

- A. A portfolio that looks diversified based on calm-market history can behave like a concentrated bet exactly when a crisis hits
- B. Correlation has no real-world consequences for portfolios
- C. It only matters for academic research papers
- D. Correlation is purely a theoretical concept with no practical use

5. What does this lesson recommend when assessing a portfolio's risk?

- A. Only ever use calm-market correlation assumptions
- B. Ignore correlation entirely and focus only on individual asset returns
- C. Stress-test the portfolio under both calm and crisis correlation assumptions
- D. Assume all assets are always perfectly correlated

Answer Key

1. What does this lesson mean by correlations being "regime-dependent"?

Correct answer: C. Correlation relationships between assets can shift depending on whether markets are calm or stressed

Correlation relationships aren't fixed - they shift depending on the market regime.

2. What typically happens to most cross-asset correlations during a genuine crisis?

Correct answer: B. They tend to rise toward +1, as a broad sell-everything dynamic takes hold

A broad 'risk-off' panic tends to push correlations across many asset classes toward +1.

3. In the illustrative example, what was notable about the safe-haven currency's correlation to stocks?

Correct answer: D. Its negative correlation to stocks actually strengthened during the crisis

Unlike most other pairs, its negative correlation strengthened, making it valuable as a hedge.

4. Why is cross-asset correlation described as a real risk-management concern?

Correct answer: A. A portfolio that looks diversified based on calm-market history can behave like a concentrated bet exactly when a crisis hits

A portfolio diversified based on calm-market correlations can behave like a concentrated bet in a crisis.

5. What does this lesson recommend when assessing a portfolio's risk?

Correct answer: C. Stress-test the portfolio under both calm and crisis correlation assumptions

A portfolio should be evaluated under both calm and crisis correlation assumptions.